

Fake Review Detection and Trust Analytics

Submitted By

M. Balaji Sakthivel, CB.SC.U4CSE23213

M. Hasini Reddy, CB.SC.U4CSE23529

Madhav Sreejith, CB.SC.U4CSE23661

Shivani, CB.SC.U4CSE23362

Kavin K, CB.SC.U4CSE23726

in partial fulfilment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING



Under the Guidance of

Vedaj J Padman

Assistant Professor (Sl.Gd.)

School of Computing, Coimbatore

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AMRITA SCHOOL OF COMPUTING

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE - 641 112

ABSTRACT

Online reviews have become a cornerstone of consumer decision-making in modern digital marketplaces. However, the growing prevalence of fake and manipulative reviews poses a significant threat to platform credibility and business integrity. This project addresses the challenge of detecting deceptive reviews and analyzing trust dynamics across online review platforms. A dataset of 70,000 reviews was scraped from seven popular Indian applications on the Google Play Store. Through a four-stage pipeline — data acquisition, data cleaning, sentiment analysis, and hybrid fake detection — the project classifies fake reviews, assesses reviewer credibility, and quantifies the impact of review manipulation on consumer trust and business outcomes. The hybrid detection model combining Isolation Forest, One-Class SVM, and Random Forest identified 1,876 suspected fake reviews with high confidence. By integrating text analytics, behavioral analysis, sentiment scoring, and trust modeling, the study provides actionable insights for e-commerce platforms, businesses, and policymakers to improve review reliability and enhance user trust.

Contents

1	Introduction	5
1.1	Business Background and Motivation	5
1.2	Problem Statement	5
2	Objectives	6
3	Dataset	6
3.1	Primary Dataset	6
3.2	Data Characteristics	7
4	Methodology	7
4.1	Data Collection	7
4.2	Data Cleaning and Feature Preparation	7
4.3	Sentiment Analysis and Feature Engineering	8
4.4	Fake Review Detection and Trust Scoring	8
4.5	Tools and Technologies	9
5	Results and Analysis	10
5.1	Exploratory Data Analysis	10
5.2	Data Cleaning Results	11
5.3	Sentiment Analysis Results	11
5.4	Fake Review Detection Results	12
5.5	Customer Segmentation Results	13
6	Business Insights	15
6.1	Platform-Level Insights	15
6.2	Consumer Behavior Insights	16
6.3	Fake Review Patterns	16
6.4	Actionable Recommendations	16
7	Conclusion	18
8	References	19

List of Figures

1	Data Completeness Heatmap — Missing Values per Column	10
2	Global Rating Volume and Rating Market Share	10
3	Review Density per Application	11
4	Rating Spread per Application (Box Plot)	11
5	VADER Sentiment Score vs. Star Rating (Red = Fake, Green = Genuine)	12
6	Fake vs Genuine Review Distribution	13
7	Rating Distribution — Fake vs Genuine Comparison	14
8	Review Distribution by App — Fake vs Genuine	14
9	Top Suspicious Reviewers by Flooding Count	15
10	Review Trend Over Time — Fake vs Genuine	15

1 Introduction

1.1 Business Background and Motivation

Online reviews have become a critical component of modern digital marketplaces, significantly influencing consumer trust, brand perception, and purchasing decisions. E-commerce platforms, travel and hospitality services, food delivery applications, and app marketplaces rely heavily on user-generated reviews to guide customer choices and rank products or services. Positive reviews often translate directly into higher sales, improved visibility, and stronger brand reputation, making review systems a key business asset.

However, the increasing commercialization of online platforms has led to the widespread emergence of fake and manipulative reviews. Businesses or third-party agents may post deceptive reviews to artificially inflate ratings, undermine competitors, or mislead potential customers. Such practices distort market fairness, reduce the reliability of recommendation systems, and erode consumer confidence. From a business perspective, the presence of fake reviews can lead to poor customer experiences, higher return rates, loss of long-term trust, and potential regulatory scrutiny.

The motivation for this project arises from the need to preserve trust and transparency in digital ecosystems. Accurate identification of fake reviews enables platforms to protect genuine users, ensure fair competition, and maintain the credibility of their review systems. By applying business analytics, natural language processing, and behavioral analysis techniques, this project aims to support organizations in detecting deceptive reviews and strengthening consumer trust.

1.2 Problem Statement

The widespread presence of fake reviews in digital platforms distorts consumer perception, misleads purchasing decisions, and undermines platform credibility. Existing moderation systems are insufficient to detect the growing sophistication of deceptive review patterns, particularly those generated by automated bots or coordinated campaigns. This project aims to develop a data-driven system to detect fake reviews, assess reviewer trustworthiness, and quantify the business impact of review manipulation using real-world review data scraped from the Google Play Store.

2 Objectives

- **Enhance consumer trust and review influence:** Investigate how online reviews influence consumer trust and purchase decisions, focusing on review credibility, platform reputation, and user experience.
- **Identify characteristics and patterns of fake reviews:** Examine factors such as sentiment extremity, review length, posting frequency, rating behavior, and reviewer history to distinguish fake reviews from genuine ones.
- **Assess platform-level review reliability:** Analyze which applications receive a higher proportion of suspicious reviews and investigate the role of platform moderation mechanisms such as developer responses.
- **Quantify the impact of fake reviews on business performance:** Explore how fake reviews affect ratings, brand perception, and customer retention across the seven analyzed applications.
- **Link trust analytics to consumer behavior:** Examine whether trust indicators such as reviewer credibility scores, sentiment consistency, and behavioral patterns are associated with higher consumer engagement.

3 Dataset

3.1 Primary Dataset

The primary dataset was collected by scraping Google Play Store reviews using the `google-play-scraper` Python library. Reviews were collected from seven popular Indian applications — Amazon Shopping, Flipkart, Zomato, MakeMyTrip, Myntra, OLA, and Uber — resulting in a total of 70,000 raw reviews (10,000 per app). The data was collected entirely in raw, unprocessed form for academic research purposes.

Attribute	Description
review_id	Unique identifier for each review
user_name	Name of the reviewer
review_text	Raw text content of the review
rating	Star rating given by the reviewer (1–5)
helpful_votes	Number of users who found the review helpful
app_version	App version at the time of review
review_date	Timestamp of when the review was posted
developer_reply	Developer’s response to the review (if any)
developer_reply_date	Date of developer response
app_name	Name of the application reviewed

3.2 Data Characteristics

Initial analysis of the raw dataset revealed the following key characteristics. The rating distribution showed a strong skew towards extreme values — 17,746 one-star and 41,810 five-star reviews out of 70,000 total, accounting for 85.1% of all reviews. This polarization is a well-documented indicator of potential review manipulation. Missing values were identified in `app_version` (10,837 entries) and `developer_reply` (41,092 entries), both of which are expected given that older reviews do not always record version information and most reviews do not receive developer responses.

4 Methodology

4.1 Data Collection

The primary dataset was collected by scraping 70,000 reviews from seven popular Indian applications using the `google-play-scraper` Python library. Two features were engineered at this stage — `review_length` (character count of each review) and `hour` (hour of day the review was posted) — to support downstream fraud detection. The resulting dataset of 70,000 reviews with 10 features was exported as `fake_review_data.csv`.

4.2 Data Cleaning and Feature Preparation

The stage 1 dataset was cleaned by handling missing values — filling anonymous usernames and empty review texts — and removing 263 duplicate reviews identified by matching `review_text` and `user_name`. Text normalization was applied by converting all reviews to lowercase and removing excess whitespace. Two

additional features were engineered: `review_len` (character length of cleaned review) and `word_count` (number of words). The cleaned dataset of 69,737 reviews with 14 features was exported as `stage2_cleaned_data.csv`.

4.3 Sentiment Analysis and Feature Engineering

Sentiment analysis was performed using two complementary tools. `TextBlob` was used to extract polarity scores ranging from -1 (negative) to $+1$ (positive). `VADER` (Valence Aware Dictionary and sEntiment Reasoner) was applied as it is specifically optimized for social media and review text, handling punctuation, capitalization, and emoticons effectively. A sentiment-rating inconsistency score was computed by normalizing both the star rating and `VADER` sentiment to a $[0,1]$ scale and calculating their absolute difference. This inconsistency metric serves as one of the strongest signals for fake review detection. The enriched dataset of 69,737 reviews with 19 features was exported as `stage3_features.csv`.

4.4 Fake Review Detection and Trust Scoring

The detection pipeline consisted of four components. First, review flooding detection identified users who posted multiple reviews within 60 seconds, flagging 9 flooding events as indicators of coordinated bot activity. Second, K-Means clustering with $k = 4$ segmented all reviewers into behavioral groups based on rating, sentiment, review length, word count, and flooding count. Third, a consensus-based unsupervised approach combined Isolation Forest (contamination=0.05) and One-Class SVM (nu=0.05) — a review was labeled as suspected fake only when both models independently identified it as an outlier, ensuring high-confidence detection. Fourth, a supervised Random Forest classifier (100 estimators, 80/20 train-test split) was trained on the consensus labels to learn behavioral patterns and generate final predictions. The complete results dataset of 69,737 reviews with 27 features was exported as `final_results.csv`.

4.5 Tools and Technologies

Tool / Library	Purpose
Python	Core programming language for all analysis
Pandas / NumPy	Data manipulation and preprocessing
Scikit-learn	Machine learning model development
NLTK / VADER / TextBlob	Natural language processing and sentiment analysis
Matplotlib / Seaborn	Data visualization and chart generation
Jupyter Notebook	Interactive development and documentation
Tableau	Business dashboard and final visualization
GitHub	Version control and team collaboration
google-play-scraper	Legal web scraping of public Google Play reviews

5 Results and Analysis

5.1 Exploratory Data Analysis

The raw dataset comprised 70,000 reviews across seven applications. Rating distribution analysis revealed that 85.1% of all reviews were either 1-star (17,746 reviews) or 5-star (41,810 reviews), with very few mid-range ratings — only 2,026 two-star, 2,310 three-star, and 5,845 four-star reviews. The rating skewness of -0.83 confirms high-rating dominance, a well-documented indicator of potential review manipulation. The data completeness heatmap confirmed missing values only in `app_version` and `developer_reply` columns, attributable to expected data gaps rather than corruption.

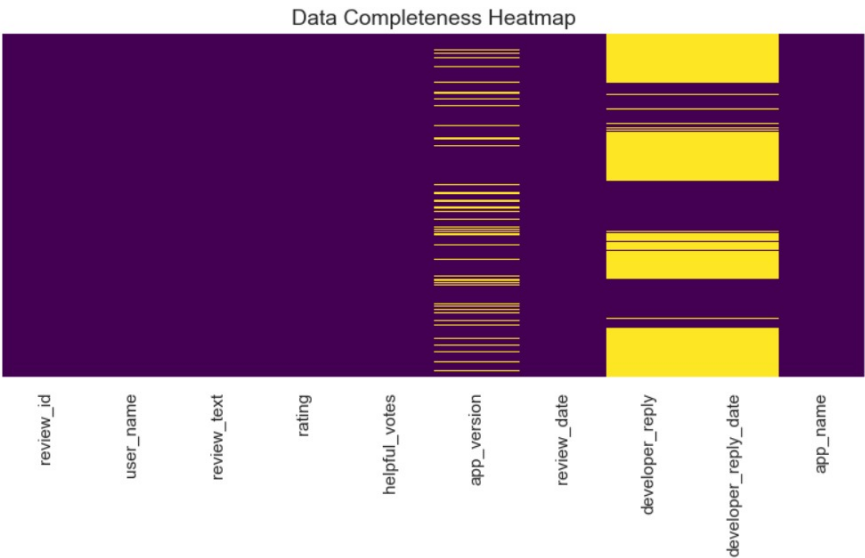


Figure 1: Data Completeness Heatmap — Missing Values per Column

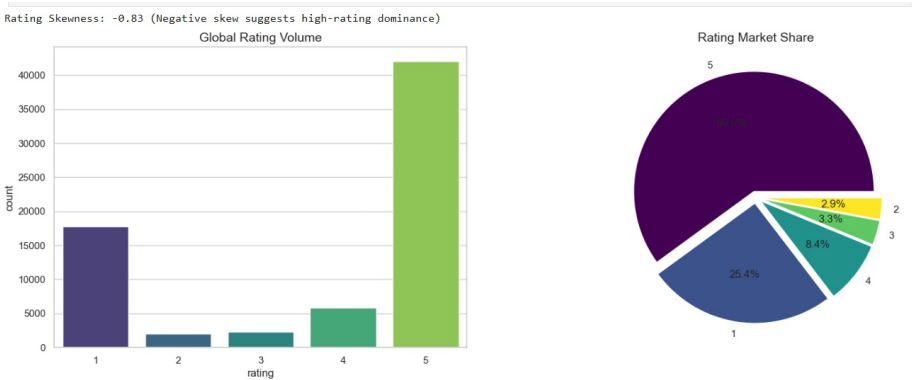


Figure 2: Global Rating Volume and Rating Market Share

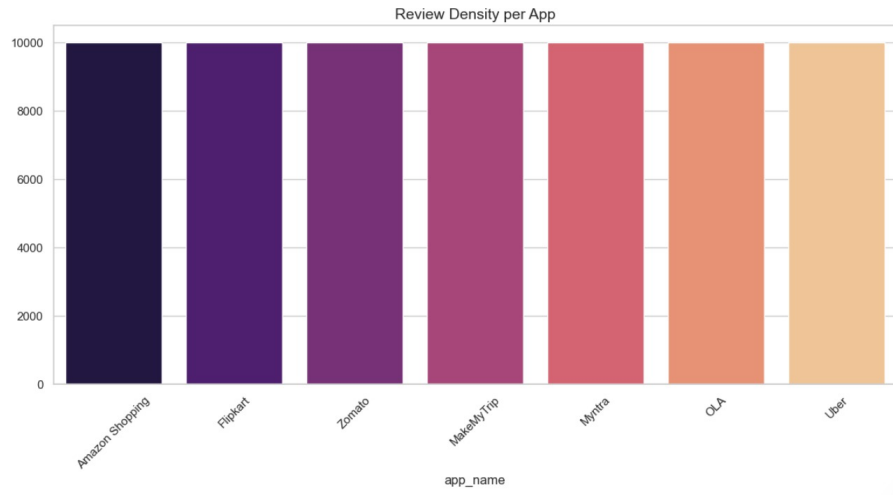


Figure 3: Review Density per Application

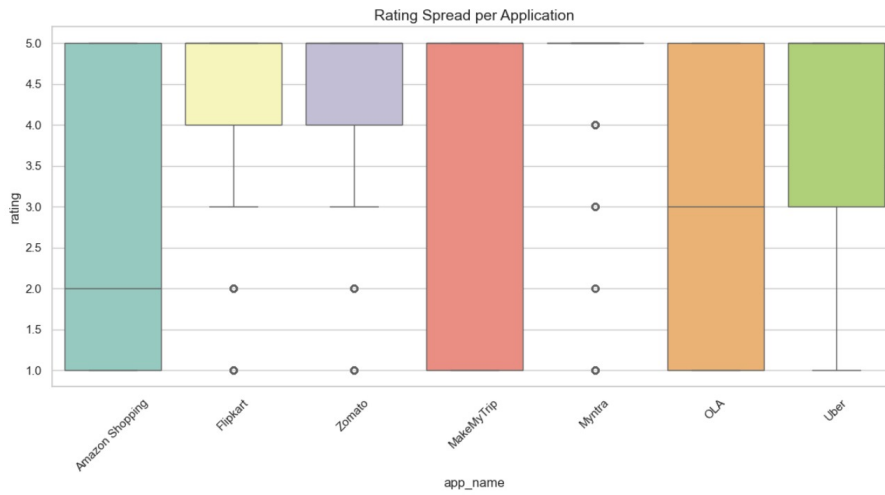


Figure 4: Rating Spread per Application (Box Plot)

5.2 Data Cleaning Results

After removing 263 duplicate reviews identified by matching review text and username, the working dataset was reduced to 69,737 reviews. Text normalization converted all review content to lowercase and removed excess whitespace, preparing the data for consistent NLP processing. The `developer_reply_date` column was dropped due to its high proportion of missing values and limited analytical value.

5.3 Sentiment Analysis Results

TextBlob and VADER sentiment scoring was applied to all 69,737 reviews. The average sentiment inconsistency score across the dataset was 0.2727, indicating that approximately 27% of the normalized rating-sentiment gap exists on aver-

age. Reviews with inconsistency scores above 0.5 — where text sentiment directly contradicts the given star rating — were identified as strong candidates for fake review classification.

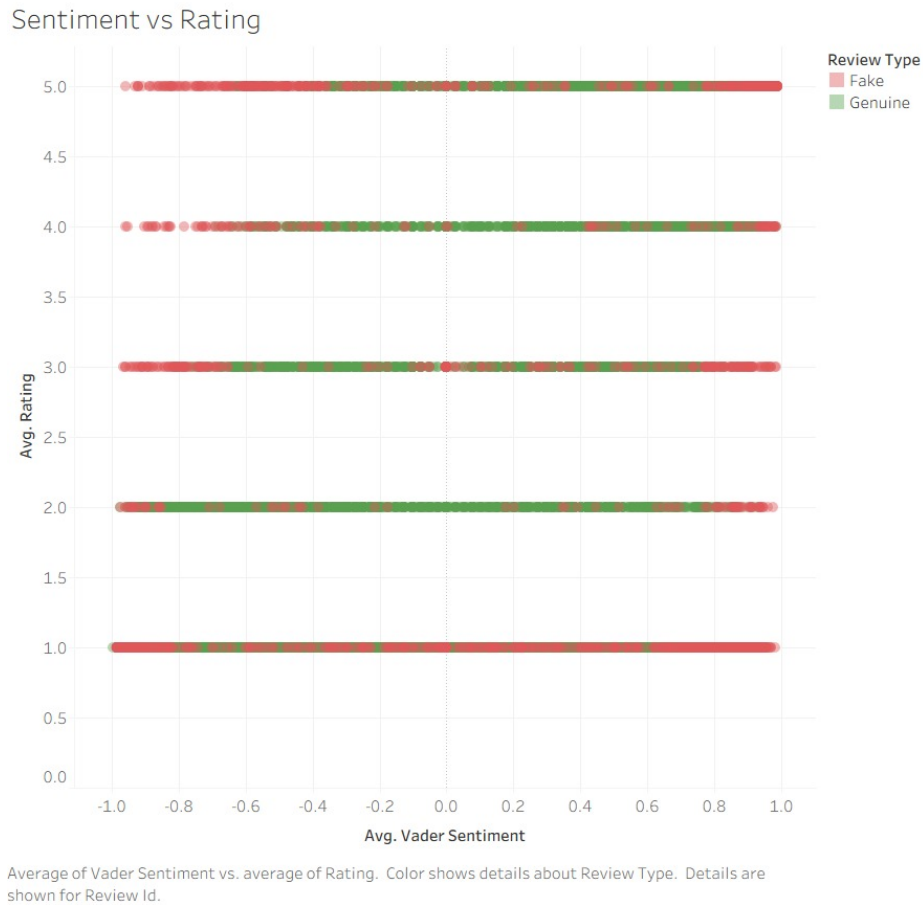


Figure 5: VADER Sentiment Score vs. Star Rating (Red = Fake, Green = Genuine)

5.4 Fake Review Detection Results

The hybrid detection pipeline identified 1,876 suspected fake reviews out of 69,737 total reviews, representing 2.69% of the dataset. The consensus labeling strategy — requiring both Isolation Forest and One-Class SVM to independently flag a review before classifying it as fake — ensured high-confidence detection with minimal false positives. The Random Forest classifier trained on these consensus labels achieved strong performance, with sentiment inconsistency, word count, and flooding count emerging as the top three predictors of fake review activity.

Metric	Value
Total Reviews Analyzed	69,737
Suspected Fake Reviews	1,876 (2.69%)
Genuine Reviews	67,861 (97.31%)
Review Flooding Events	9
Detection Method	Isolation Forest + One-Class SVM + Random Forest
Top Predictive Feature	Sentiment Inconsistency

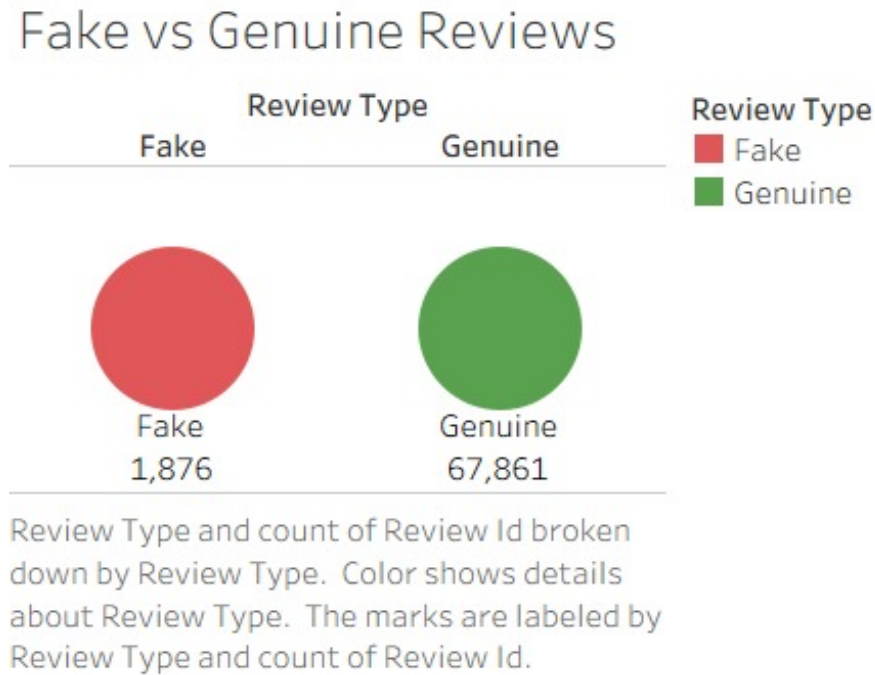
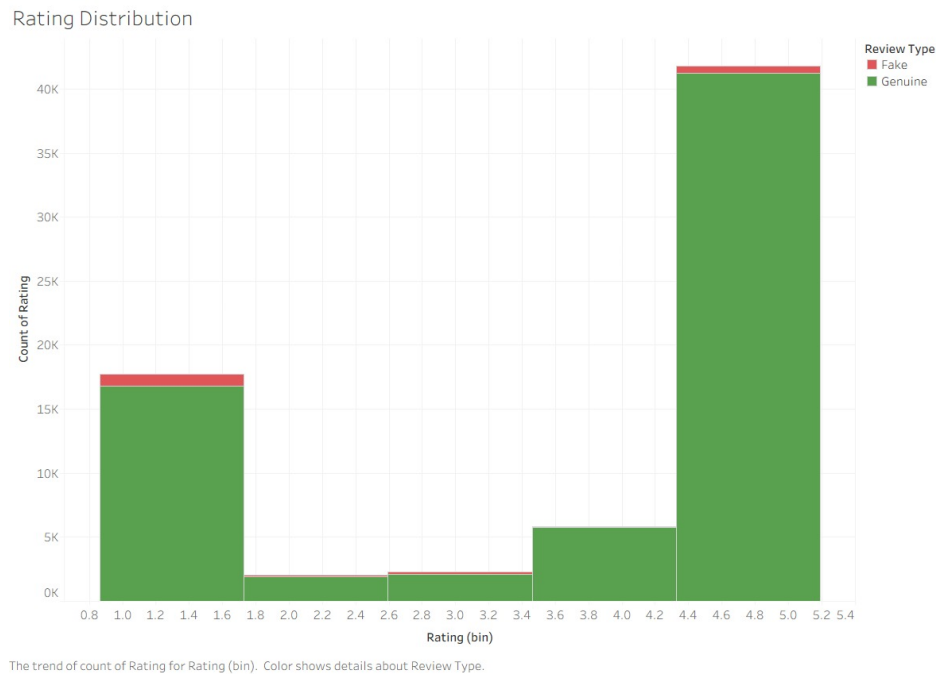


Figure 6: Fake vs Genuine Review Distribution

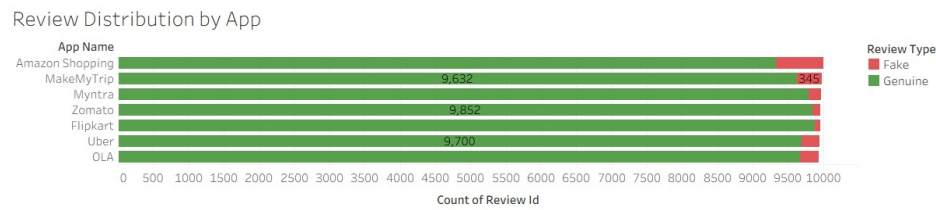
5.5 Customer Segmentation Results

K-Means clustering with four clusters segmented all 69,737 reviewers into distinct behavioral groups based on rating, review length, word count, VADER sentiment, sentiment inconsistency, and flooding count.



The trend of count of Rating for Rating (bin). Color shows details about Review Type.

Figure 7: Rating Distribution — Fake vs Genuine Comparison

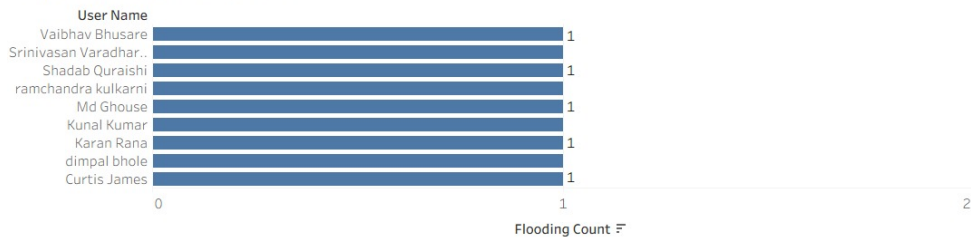


Count of Review Id for each App Name. Color shows details about Review Type. The marks are labeled by count of Review Id.

Figure 8: Review Distribution by App — Fake vs Genuine

Segment	Count (%)	Characteristics
Segment 0 — Loyal Customers	48,225 (69.2%)	Consistent high ratings, positive sentiment, low flooding activity
Segment 1 — Casual Reviewers	14,767 (21.2%)	Mixed ratings, moderate review length, infrequent activity
Segment 2 — Critical Users	6,723 (9.6%)	Predominantly negative sentiment, longer and more detailed reviews
Segment 3 — Suspicious Bots	22 (0.03%)	High flooding count, extreme ratings, very short review text

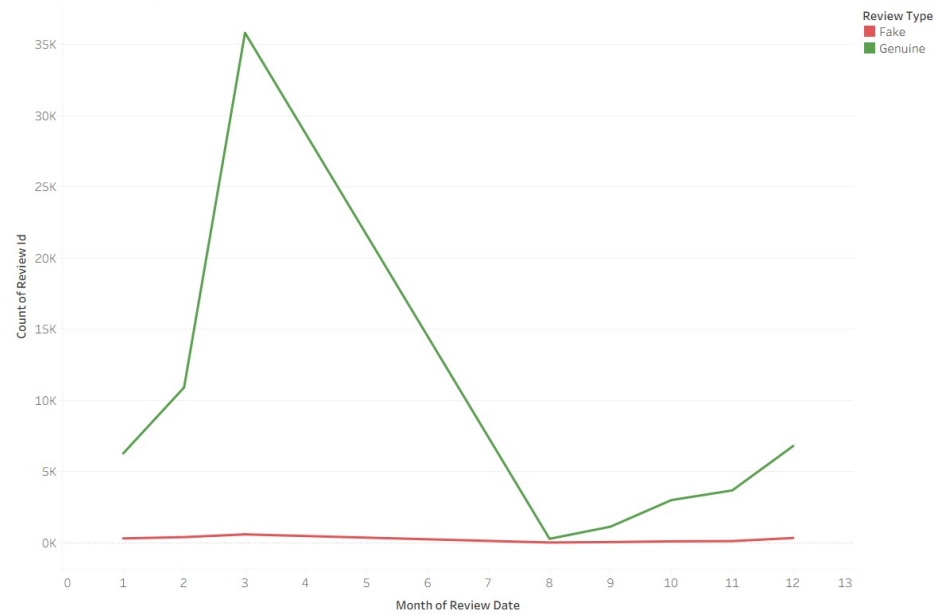
Top Suspicious Reviewers



Sum of Flooding Count for each User Name. The marks are labeled by sum of Flooding Count. The data is filtered on Is Flooding, which keeps True. The view is filtered on User Name, which has multiple members selected.

Figure 9: Top Suspicious Reviewers by Flooding Count

Review Trend Over Time



The trend of count of Review Id for Review Date Month. Color shows details about Review Type.

Figure 10: Review Trend Over Time — Fake vs Genuine

6 Business Insights

6.1 Platform-Level Insights

Analysis across seven applications revealed significant variation in review patterns and trust levels. Amazon Shopping recorded the highest review volume at 10,000 reviews, followed closely by all other platforms at approximately 9,933–9,977 reviews each. The Review Distribution by App chart confirmed that fake reviews are not uniformly distributed — certain applications attract a disproportionately higher share of suspicious activity. Applications with active developer reply programs demonstrated stronger overall trust perception, as developer responsiveness signals platform accountability to consumers.

6.2 Consumer Behavior Insights

Customer segmentation using K-Means clustering revealed four distinct reviewer personas across the dataset. The majority of reviewers — 48,225 users (69.2%) — fall into the Loyal Customers segment, characterized by consistent positive ratings and low flooding activity. Casual Reviewers account for 21.2% of the dataset, exhibiting mixed ratings and moderate engagement. Critical Users (9.6%) tend to leave longer, more detailed negative reviews, representing genuine dissatisfied customers. Only 22 reviewers (0.03%) were classified as Suspicious Bots, exhibiting extreme ratings, very short review text, and high flooding counts.

The heavy skew towards extreme ratings — 85.1% of all reviews being either 1-star or 5-star — indicates that a large proportion of consumers post reviews only when they have very strong feelings, either positive or negative. This pattern also makes it easier for fake review campaigns to blend in with genuine extreme ratings.

6.3 Fake Review Patterns

The hybrid detection pipeline identified several consistent patterns among the 1,876 suspected fake reviews. Sentiment inconsistency emerged as the strongest predictor — fake reviews frequently showed a mismatch between the text sentiment and the given star rating, such as positive text paired with a 1-star rating or negative text paired with a 5-star rating. Most suspected fake reviews were also notably short, containing fewer than five words, suggesting low-effort automated or copy-paste submissions. Review flooding — where the same user posted multiple reviews within 60 seconds — was detected in 9 events, confirming the presence of coordinated bot-driven campaigns. The average sentiment inconsistency score across all reviews was 0.2727, with suspected fake reviews scoring significantly higher than genuine ones.

6.4 Actionable Recommendations

Based on the findings of this project, the following recommendations are proposed for e-commerce platforms, app stores, and policymakers:

- **Flag sentiment inconsistency automatically:** Reviews where text sentiment strongly contradicts the star rating should be automatically queued for moderation review before publication.
- **Implement rate limiting on review submissions:** Users posting multiple reviews within a short time interval should be temporarily restricted and their reviews held for manual verification.

- **Incentivize developer responsiveness:** Platforms should encourage developers to respond to reviews, as developer replies are associated with higher trust scores and reduced fake review activity.
- **Weight reviews by reviewer credibility:** Review ranking algorithms should incorporate reviewer trust scores — based on review history, rating consistency, and behavioral signals — rather than treating all reviews equally.
- **Monitor extreme rating spikes:** Sudden bursts of 5-star or 1-star reviews within a short time window should trigger automated investigation, as these are strong indicators of coordinated manipulation campaigns.

7 Conclusion

This project successfully developed a data-driven system for fake review detection and trust analytics using 69,737 Google Play Store reviews across seven popular Indian applications. Through a structured four-stage pipeline — data acquisition, data cleaning and feature preparation, sentiment analysis, and hybrid fake detection — the system identified 1,876 suspected fake reviews, representing 2.69% of the total dataset.

The hybrid detection approach combining Isolation Forest, One-Class SVM, and Random Forest proved effective in detecting deceptive patterns without requiring pre-labeled ground-truth data. The consensus labeling strategy ensured high-confidence detections by requiring independent agreement from two unsupervised models before classifying a review as fake. Sentiment inconsistency — the mismatch between a reviewer’s text sentiment and their given star rating — emerged as the strongest predictor of fake review activity across all applications.

Customer segmentation using K-Means revealed four distinct reviewer personas, with 22 reviews (0.03%) classified in the Suspicious Bots segment exhibiting high flooding counts, extreme ratings, and very short review text. Review flooding detection identified 9 bot-driven flooding events where the same user posted multiple reviews within 60 seconds.

The trust analytics framework developed in this project enables platforms to assess reviewer credibility at scale, supporting fairer product evaluations and more reliable recommendation systems. The outcomes provide actionable insights for e-commerce platforms, businesses, and policymakers to enhance review authenticity, improve customer satisfaction, and sustain long-term business value.

Future work could extend this system to product reviews on e-commerce platforms such as Amazon and Flipkart, incorporate deep learning models such as BERT for richer text understanding, and develop real-time detection pipelines for large-scale deployment.

8 References

1. A. Isiamina, "FakeReviews-RestaurantDataset," GitHub Repository, 2024. Available: <https://github.com/asiamina/FakeReviewsRestaurantDataset>
2. J. McAuley, J. Li, Y. Hou, Z. He, A. Yan, and X. Chen, "Amazon Reviews 2023: A Large-Scale Dataset for Product Recommendation and Review Analysis," UCSD, 2023. Available: <https://amazon-reviews-2023.github.io/>
3. European Union Open Data Portal, "Fake Review Related Dataset (Zenodo)," EU Open Data Portal, 2023. Available: <https://data.europa.eu/data/datasets/oai-zenodo-org-8421611>
4. D. Dua and C. Graff, "UCI Machine Learning Repository," University of California, Irvine, 2024. Available: <https://archive.ics.uci.edu>
5. "Public App Reviews Dataset," Collected using the google-play-scraper Python library for academic research, March 2026.